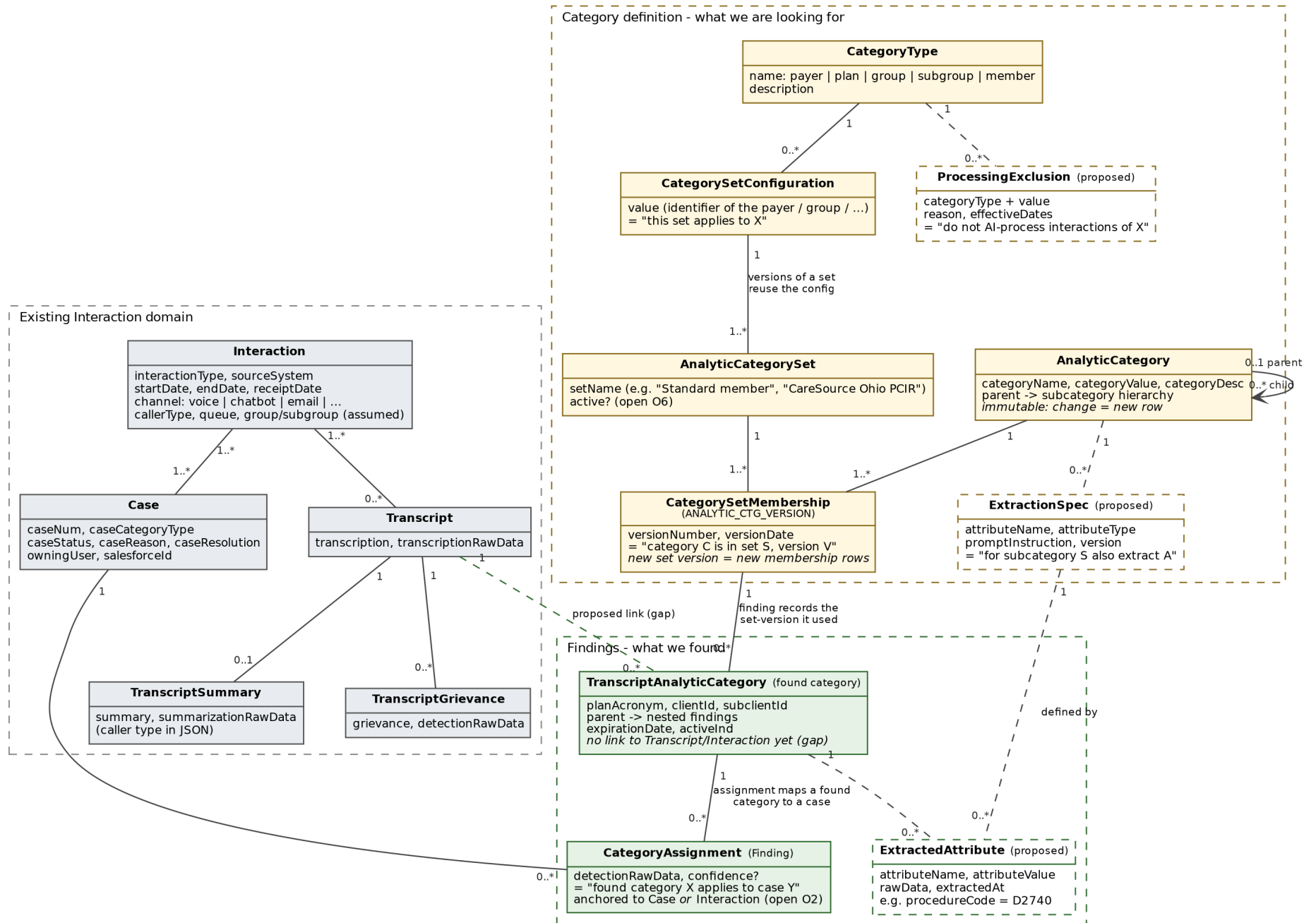


Call Analysis - UML Domain Model (draft)

Solid boxes = concepts already in the ER diagram | Dashed boxes = proposed additions | Grey = existing Interaction domain



Concept definitions

Concept	Meaning	ER diagram table
Interaction	One contact with a customer on any channel (voice call today; chatbot, e-mail later). Carries caller type, queue, source and - by working assumption - group/subgroup.	INTERACTION
Case	A unit of work opened for an interaction. Many-to-many with Interaction in the model; implemented closer to one-to-many. One interaction may open several cases (provider calling about several members).	CASE, INTERACTION_CASE
Transcript / Summary / Grievance	Output of the existing Call Summarization service; the input to Call Analysis.	TRANSCRIPT, TRANSCRIPT_SUMMARY, TRANSCRIPT_GRIEVANCE
CategoryType	The level a category set can be applied at: payer, plan, group, subgroup, member (extensible, e.g. TPA).	CATEGORY_TYPE
CategorySetConfiguration	Binds a set to a specific instance of a level via a type + identifier value (e.g. payer = CareSource Ohio). Stable across versions of the same set.	CATEGORY_SET_CONF
AnalyticCategorySet	A named collection of categories used to classify interactions (e.g. Standard member set, CareSource Ohio PCIR). The set row is stable; its composition is versioned through membership rows.	ANALYTIC_CATEGORY_SET
AnalyticCategory	A single category or subcategory (via parent). Immutable: renaming or redefining creates a new row. By convention belongs to exactly one set (enforced via membership, not a column).	ANALYTIC_CATEGORY
CategorySetMembership	Records that a category is in a set at version V (version number and date live here). A new set version = a new group of membership rows. Findings point here, so every finding knows the exact set version it was classified under.	ANALYTIC_CTG_VERSION
TranscriptAnalyticCategory	A found category: the AI detected category X in a transcript, under set-version V, stamped with the plan/client/subclient observed on the interaction. Parent link allows nested findings.	TRANSCRIPT_ANALYTIC_CTG
CategoryAssignment (Finding)	Maps a found category to a case (0..* assignments per found category). This is where the multi-case mapping problem lands.	CATEGORY_ASSIGNMENT
ExtractionSpec (proposed)	Per-subcategory definition of the additional attributes to extract (the drill-down). Versioned so reprocessing can be tied to a spec version.	- (none yet) - resolves open item O1
ExtractedAttribute (proposed)	A name/value extracted for a found category under an ExtractionSpec (e.g. procedureCode = D2740, deliveryMechanism = email).	- (none yet) - resolves open item O1
ProcessingExclusion (proposed)	Opt-out rule: interactions belonging to a given type + value are not AI-processed.	- (none yet) - feature F21

Where the model and the ER diagram differ (to validate)

#	Difference / question	Table(s)
1	TRANSCRIPT_ANALYTIC_CTG has no foreign key to TRANSCRIPT or INTERACTION; it is reachable only through CATEGORY_ASSIGNMENT -> CASE -> INTERACTION. If a found category cannot be mapped to a case (or anchoring moves to Interaction, O2), it becomes orphaned. Proposed: add transcript_id (or interaction_id).	TRANSCRIPT_ANALYTIC_CTG
2	CATEGORY_ASSIGNMENT references case_id; the model keeps Interaction as the alternative anchor. Decide (O2) - multi-case interactions need an extra LLM pass if Case is kept.	CATEGORY_ASSIGNMENT
3	No home for per-subcategory extracted attributes. Two proposed tables (ExtractionSpec, ExtractedAttribute) or a JSON column on TRANSCRIPT_ANALYTIC_CTG / ctg_detection_raw_data.	TRANSCRIPT_ANALYTIC_CTG

#	Difference / question	Table(s)
4	Versioning: set row is stable, version_number/version_date sit on ANALYTIC_CTG_VERSION. Confirm the rule - a new version writes a full new group of membership rows (all categories in the new version), not only the delta - and that classification uses the highest version (or an active flag, O6).	ANALYTIC_CTG_VERSION / ANALYTIC_CATEGORY_SET
5	"Category belongs to exactly one set" is convention only; nothing in the schema prevents one category appearing in two sets' membership rows. Either add a set FK on ANALYTIC_CATEGORY or enforce it in the loader.	ANALYTIC_CATEGORY
6	A category's parent (subcategory hierarchy) is not constrained to the same set; add a check or document the rule.	ANALYTIC_CATEGORY
7	A set points to exactly one CATEGORY_SET_CONF (one type + one value). Clarify how the standard set that applies to all interactions is expressed (null/wildcard value?) and how one set is applied to several groups without duplicating the set.	CATEGORY_SET_CONF
8	TRANSCRIPT_ANALYTIC_CTG carries plan_acronym / client / subclient identifiers separately from CATEGORY_SET_CONF; these are observed identifiers on the interaction, not configured applicability - keep both but name them distinctly. Its expiration_date / active_ind look like configuration fields on a findings table; confirm intent.	TRANSCRIPT_ANALYTIC_CTG
9	No exclusion/opt-out table for plans that refuse AI processing.	-
10	Interaction to Case is modelled many-to-many but implemented as one-to-many; confirm with the feature team before relying on either.	INTERACTION_CASE